

Personal Finance Simulator

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Overview

This project aims to develop a **Command-Line Interface (CLI) based Personal Finance Simulator** that helps users manage their **financial activities** such as income, expenses, budgeting, and account management.

The system will be implemented primarily using Python, focusing on core programming concepts such as **object-oriented programming (OOP), loops, conditionals, and data handling**. **PostgreSQL** will be integrated for basic data storage and persistence.

All interactions will occur through a terminal interface, emphasizing backend logic, program structure, and real-world financial simulation.

Objectives

1. Build a fully functional CLI-based **finance management system**
2. Apply Python concepts (**OOP, loops, conditionals, functions**)
3. Simulate real-world financial scenarios
4. Implement basic database operations using **PostgreSQL**
5. Ensure efficient **data handling** and modular program design

Key Features

Account Management

- Create and manage multiple account types (Savings, Checking)
- View account details and balances
- Update account information

Transaction System

- Deposit and withdraw funds
- Transfer money between accounts
- Validate transactions (e.g., insufficient balance)

Budget Management

- Set monthly budgets
- Track expenses against budgets
- Display warnings when limits are exceeded

Expense Categorization

- Categorize expenses (food, rent, travel, etc.)
- Analyze spending patterns by category

Financial Reports

- Generate summaries of income and expenses
- View transaction history

Transaction Filtering

- Filter transactions by date, type, or category
- Search for specific transactions

Technologies Used

- **Language:** Python
- **Database:** PostgreSQL
- **Library:** psycopg2 (for database connection)
- **Interface:** Command-Line Interface (CLI)

FlowChart

